

# Derivation Attempt v1:

## 5D $\rightarrow$ 4D Newton Constant

Actual Calculation (BLOCK-003)

EDC Research Program

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Status: *WORKING* / *OPEN*

### Abstract

This document attempts an explicit derivation of the effective 4D Newton constant  $G_N$  from the 5D bulk action used in EDC. We linearize around a flat background, apply Israel junction conditions at the brane, solve for the static weak-field potential, and extract  $G_N$ . The derivation succeeds in obtaining a standard brane-world result:  $G_N = \kappa_5^2/(8\pi L)$  for a compact extra dimension of size  $L$ , or  $G_N \sim \kappa_5^4 \sigma$  for RS-type warped geometries. However, the EDC-specific scaling involving powers 12/13 does *not* emerge from this standard calculation. We identify exactly where additional EDC-specific input would be required, marking BLOCK-003 as still **[OPEN]**.

## 1 Setup and Notation

### Epistemic Status

**Status:** **[BL]**/**[I]**. Standard brane-world setup with EDC coordinate conventions.

**Coordinates and signature [BL].** We use a 5D bulk manifold  $\mathcal{M}^5$  with coordinates

$$X^A = (x^\mu, \xi), \quad A = 0, 1, 2, 3, 5, \quad \mu = 0, 1, 2, 3, \quad (1)$$

where  $x^\mu = (t, x^i)$  are 4D coordinates and  $\xi$  is the extra-dimensional coordinate. Signature convention:  $(-, +, +, +, +)$ .

**Brane location [I].** A 3-brane  $\Sigma$  is located at  $\xi = 0$ . Brane observers experience the induced metric  $\gamma_{\mu\nu} = g_{\mu\nu}|_{\xi=0}$ .

**5D gravitational coupling [BL].** The fundamental coupling is  $\kappa_5^2 = 8\pi G_5$ , where  $G_5$  has dimensions  $[\text{length}]^3$ .

**Action [BL].** The total action is:

$$S = S_{\text{bulk}} + S_{\text{GHY}} + S_{\text{brane}}, \quad (2)$$

with

$$S_{\text{bulk}} = \frac{1}{2\kappa_5^2} \int_{\mathcal{M}^5} d^5 X \sqrt{-g} \left( R^{(5)} - 2\Lambda_5 \right), \quad (3)$$

$$S_{\text{GHY}} = \frac{1}{\kappa_5^2} \oint_{\partial\mathcal{M}} d^4 x \sqrt{|h|} K, \quad (4)$$

$$S_{\text{brane}} = -\sigma \int_{\Sigma} d^4 x \sqrt{-\gamma} + S_{\text{matter}}[\gamma, \psi]. \quad (5)$$

Here  $\sigma$  is the brane tension and  $K = h^{ab} K_{ab}$  is the trace of the extrinsic curvature [3, 4].

**Newtonian limit definition [BL].** The “Newtonian limit” means: static, weak-field, non-relativistic source. On the brane, we seek a potential  $\Phi(r)$  such that the 4D metric is

$$ds_{\text{brane}}^2 \approx -(1 + 2\Phi) dt^2 + (1 - 2\Phi) d\vec{x}^2, \quad (6)$$

with  $|\Phi| \ll 1$  and  $\Phi(r) \rightarrow -G_N M/r$  at large  $r$  for a point mass  $M$ .

## 2 Background and Linearization

### Epistemic Status

**Status:** [P]/[BL]. We choose a flat background (no warp) with  $\Lambda_5 = 0$  for simplicity. This is a *postulate* that limits generality.

**Background choice [P].** We take the simplest case: flat 5D Minkowski bulk with  $\Lambda_5 = 0$ :

$$\bar{g}_{AB} = \eta_{AB} = \text{diag}(-1, +1, +1, +1, +1). \quad (7)$$

The brane sits at  $\xi = 0$  and we assume  $\mathbb{Z}_2$  orbifold symmetry  $\xi \rightarrow -\xi$ . The brane tension  $\sigma$  is set to zero for the background (or treated as a first-order perturbation).

**Perturbation [BL].** Write

$$g_{AB} = \eta_{AB} + h_{AB}, \quad |h_{AB}| \ll 1. \quad (8)$$

**Gauge choice [BL].** We use the transverse-traceless (TT) gauge adapted to the brane:

$$\partial^A h_{AB} = \frac{1}{2} \partial_B h, \quad h \equiv \eta^{AB} h_{AB}. \quad (9)$$

This is the 5D analog of de Donder (harmonic) gauge.

**Linearized Einstein equations [Dc].** The 5D Einstein tensor linearized around flat space is [1, 2]:

$$G_{AB}^{(1)} = -\frac{1}{2} \square_5 \bar{h}_{AB}, \quad (10)$$

where  $\bar{h}_{AB} = h_{AB} - \frac{1}{2} \eta_{AB} h$  is the trace-reversed perturbation and

$$\square_5 = \eta^{CD} \partial_C \partial_D = -\partial_t^2 + \nabla_3^2 + \partial_\xi^2. \quad (11)$$

In our gauge, the linearized vacuum field equations become:

$$\square_5 \bar{h}_{AB} = 0 \quad (\text{bulk, away from brane}). \quad (12)$$

## 3 Israel Junction Conditions

### Epistemic Status

**Status:** [BL]. Standard Israel junction conditions [5].

**Extrinsic curvature [BL].** For a hypersurface at  $\xi = 0$  with unit normal  $n^A = (0, 0, 0, 0, 1)$ , the extrinsic curvature is:

$$K_{\mu\nu} = -\frac{1}{2}\partial_\xi g_{\mu\nu}|_{\xi=0}. \quad (13)$$

With  $\mathbb{Z}_2$  symmetry, the jump across the brane is  $[K_{\mu\nu}] = 2K_{\mu\nu}^+$  where  $K^+$  is evaluated from  $\xi > 0$ .

**Israel junction conditions [BL].** The discontinuity in extrinsic curvature is related to the brane stress-energy [5]:

$$[K_{\mu\nu}] - [K]\gamma_{\mu\nu} = -\kappa_5^2 S_{\mu\nu}, \quad (14)$$

where  $S_{\mu\nu}$  is the brane stress-energy tensor. For a brane with tension  $\sigma$  and localized matter  $\tau_{\mu\nu}$ :

$$S_{\mu\nu} = -\sigma\gamma_{\mu\nu} + \tau_{\mu\nu}. \quad (15)$$

Substituting:

$$[K_{\mu\nu}] - [K]\gamma_{\mu\nu} = \kappa_5^2 \sigma \gamma_{\mu\nu} - \kappa_5^2 \tau_{\mu\nu}. \quad (16)$$

**Linearized junction conditions [Dc].** At linear order with  $\mathbb{Z}_2$  symmetry, using (13):

$$\partial_\xi h_{\mu\nu}|_{\xi=0^+} = -\kappa_5^2 \left( \tau_{\mu\nu} - \frac{1}{3}\tau\eta_{\mu\nu} \right), \quad (17)$$

where  $\tau = \eta^{\mu\nu}\tau_{\mu\nu}$  and we have used that for a  $\mathbb{Z}_2$  brane,  $[K_{\mu\nu}] = 2K_{\mu\nu}^+ = -\partial_\xi h_{\mu\nu}|_{0^+}$ .

## 4 Static Potential on the Brane

### Epistemic Status

**Status:** [Dc]/[OPEN]. We derive the Green's function structure but encounter a divergence requiring regularization.

**Point mass source [BL].** Place a static point mass  $M$  at the origin on the brane:

$$\tau_{00} = M\delta^{(3)}(\vec{x}), \quad \tau_{ij} = 0, \quad \tau = -M\delta^{(3)}(\vec{x}). \quad (18)$$

**Mode expansion [Dc].** For the static case ( $\partial_t = 0$ ), the bulk equation (12) becomes:

$$(\nabla_3^2 + \partial_\xi^2) \bar{h}_{\mu\nu} = 0. \quad (19)$$

Fourier transform in the 3D spatial coordinates:

$$\bar{h}_{\mu\nu}(\vec{x}, \xi) = \int \frac{d^3k}{(2\pi)^3} \tilde{h}_{\mu\nu}(\vec{k}, \xi) e^{i\vec{k}\cdot\vec{x}}. \quad (20)$$

The bulk equation becomes:

$$(\partial_\xi^2 - k^2) \tilde{h}_{\mu\nu} = 0, \quad k = |\vec{k}|. \quad (21)$$

Solution with  $\mathbb{Z}_2$  symmetry and decay as  $|\xi| \rightarrow \infty$ :

$$\tilde{h}_{\mu\nu}(\vec{k}, \xi) = \tilde{h}_{\mu\nu}^{(0)}(\vec{k}) e^{-k|\xi|}. \quad (22)$$

**Apply junction condition [Dc].** From (17), the derivative jump at  $\xi = 0$  gives:

$$\partial_\xi \tilde{h}_{\mu\nu}|_{0+} = -k \tilde{h}_{\mu\nu}^{(0)} = -\kappa_5^2 \left( \tilde{\tau}_{\mu\nu} - \frac{1}{3} \tilde{\tau} \eta_{\mu\nu} \right). \quad (23)$$

Thus:

$$\tilde{h}_{\mu\nu}^{(0)}(\vec{k}) = \frac{\kappa_5^2}{k} \left( \tilde{\tau}_{\mu\nu} - \frac{1}{3} \tilde{\tau} \eta_{\mu\nu} \right). \quad (24)$$

**Brane potential [Dc].** For the 00-component with the point source (18):

$$\tilde{\tau}_{00} = M, \quad \tilde{\tau} = -M. \quad (25)$$

Hence:

$$\tilde{h}_{00}^{(0)} = \frac{\kappa_5^2}{k} \left( M + \frac{M}{3} \right) = \frac{4\kappa_5^2 M}{3k}. \quad (26)$$

Inverting the Fourier transform:

$$h_{00}(\vec{x}, 0) = \frac{4\kappa_5^2 M}{3} \int \frac{d^3 k}{(2\pi)^3} \frac{e^{i\vec{k} \cdot \vec{x}}}{k}. \quad (27)$$

### Derivation Incomplete

The integral (27) is:

$$\int \frac{d^3 k}{(2\pi)^3} \frac{e^{i\vec{k} \cdot \vec{x}}}{k} = \frac{1}{2\pi^2 r^2}. \quad (28)$$

This gives  $h_{00} \sim 1/r^2$ , **not** the Newtonian  $1/r$  behavior!

**Missing element:** In a non-compact extra dimension, gravity “leaks” into the bulk, producing  $1/r^2$  at all scales. To recover 4D  $1/r$  gravity, we need either:

1. A **compact** extra dimension of size  $L$  (Kaluza–Klein), or
2. A **warped** geometry that localizes the graviton zero mode (Randall–Sundrum).

## 5 Compact Extra Dimension: Recovery of 4D Gravity

### Epistemic Status

**Status:** [P]/[Dc]. We *assume* a compact  $\xi$ -direction of size  $L$ . This is a postulate.

**Compactification [P].** Let  $\xi \in [0, L]$  with  $\mathbb{Z}_2$  orbifold at both ends (or periodic with period  $2L$ ). The bulk solution now involves a discrete mode sum:

$$\tilde{h}_{\mu\nu}(\vec{k}, \xi) = \sum_{n=0}^{\infty} \tilde{h}_{\mu\nu}^{(n)}(\vec{k}) \cos\left(\frac{n\pi\xi}{L}\right). \quad (29)$$

**Zero mode dominance [Dc].** At distances  $r \gg L$ , only the  $n = 0$  (zero) mode contributes significantly. The zero mode is constant in  $\xi$  and satisfies:

$$\nabla_3^2 \bar{h}_{\mu\nu}^{(0)} = -\frac{\kappa_5^2}{L} \left( \tau_{\mu\nu} - \frac{1}{3} \tau \eta_{\mu\nu} \right) \delta^{(3)}(\vec{x}), \quad (30)$$

where the factor  $1/L$  comes from integrating over the extra dimension.

**4D Poisson equation [Dc].** This is exactly the 4D linearized Einstein equation:

$$\nabla^2 h_{00}^{(0)} = -\frac{4\kappa_5^2 M}{3L} \delta^{(3)}(\vec{x}). \quad (31)$$

Solution:

$$h_{00}^{(0)}(r) = \frac{\kappa_5^2 M}{3\pi L} \cdot \frac{1}{r}. \quad (32)$$

**Identification of  $G_N$  [Dc].** Comparing with the Newtonian potential  $\Phi = -G_N M/r$  and using  $h_{00} = -2\Phi$ :

$$-2\Phi = \frac{\kappa_5^2 M}{3\pi L r} \Rightarrow \Phi = -\frac{\kappa_5^2 M}{6\pi L r}. \quad (33)$$

Thus:

$$G_N = \frac{\kappa_5^2}{6\pi L} = \frac{G_5}{L}. \quad (34)$$

**Result (Kaluza–Klein):**

$$G_N = \frac{\kappa_5^2}{6\pi L} = \frac{4G_5}{3L}, \quad (35)$$

where  $L$  is the compactification length.

**Epistemic status:** [Dc] derived from linearized 5D GR with compact extra dimension.

**Required input:** The length scale  $L$  must be specified independently.

## 6 Warped Geometry Alternative

### Epistemic Status

**Status:** [P]/[BL]. Standard RS result quoted; full derivation requires solving warped bulk equations.

**Randall–Sundrum setup [BL].** For a warped metric  $ds^2 = e^{-2A(\xi)} \eta_{\mu\nu} dx^\mu dx^\nu + d\xi^2$  with AdS<sub>5</sub> bulk [6]:

$$G_N = \frac{\kappa_5^4 \sigma}{48\pi} \quad (\text{RS2 limit}), \quad (36)$$

where the effective 4D Planck mass is set by the warp factor localization.

**Key difference from EDC [I].** The RS result (36) gives  $G_N \sim \kappa_5^4 \sigma$ . This is *not* the structure appearing in EDC gravitational claims, which involve specific power combinations (12, 13) of other parameters.

## 7 Confrontation with EDC Scaling

### Derivation Incomplete

**The derivation does NOT produce EDC-specific powers (12/13).**

From standard brane-world physics, we obtain:

- **Kaluza–Klein:**  $G_N = \kappa_5^2/(6\pi L)$  depends on compactification scale  $L$ .
- **Randall–Sundrum:**  $G_N \sim \kappa_5^4 \sigma$  depends on tension and 5D coupling.

Neither of these has the form claimed in EDC gravitational sector discussions. To obtain

powers 12/13 would require:

1. A specific EDC-derived relationship between  $L$  (or warp scale) and other EDC parameters ( $\sigma$ ,  $R_\xi$ , membrane geometry), **not derived here** [OPEN].
2. Or a fundamentally different mechanism beyond standard linearized brane-world gravity.

**Conclusion:** BLOCK-003 remains [OPEN]. The power-law structure is **not derived** by this calculation.

## 8 Epistemic Ledger

| Step                        | Status | Notes                                   |
|-----------------------------|--------|-----------------------------------------|
| 5D action                   | [BL]   | Standard Einstein–Hilbert + GHY + brane |
| Flat background choice      | [P]    | Simplification; limits generality       |
| Linearized equations        | [Dc]   | Derived from action                     |
| Israel junction conditions  | [BL]   | Standard result [5]                     |
| Mode expansion              | [Dc]   | Standard Fourier methods                |
| Non-compact: $1/r^2$        | [Dc]   | Derived; incorrect for 4D gravity       |
| Compactification $L$        | [P]    | Postulated to recover $1/r$             |
| $G_N = \kappa_5^2/(6\pi L)$ | [Dc]   | Derived given compactification          |
| RS warped result            | [BL]   | Quoted from literature                  |
| EDC powers 12/13            | [OPEN] | Not derived; no mechanism identified    |

## 9 What Is Required to Close BLOCK-003

Missing lemmas [OPEN].

1. **Length scale origin:** The compactification length  $L$  (or warp scale in RS) must be derived from EDC-internal parameters, not imposed externally.
2. **Power structure:** If EDC claims  $G_N \sim \sigma^a R_\xi^b$  with specific powers  $a, b$ , then either:
  - A derivation of  $L = f(\sigma, R_\xi, \dots)$  from EDC geometry, or
  - A non-standard gravitational mechanism specific to EDC.
3. **Anti-circularity:** The derivation must not use  $G_N$  as input. Currently, (34) satisfies this (uses only  $\kappa_5$ ,  $L$ ), but  $L$  itself may hide circularity if defined via 4D observations.

## 10 Separation of Concerns

This derivation attempt (v1) is a *working document* linked to the frozen BLOCK-003 program note in `paper_gravity_block003/`.

- The program note defines acceptance criteria AC-G1 through AC-G10.
- This document addresses AC-G1 (action), AC-G2 (junction), AC-G3 (background), AC-G4 (linearization), AC-G6 (propagator structure), and partially AC-G7 ( $G_N$  extraction).

- AC-G5 (KK modes), AC-G8 (numerical consistency), AC-G9 (parameter counting), AC-G10 (reproducibility) remain open or partially addressed.

The nuclear pinning monograph (`edc_book_2/`) is **not modified** by this work.

## 11 Conclusion

1. We derived the standard brane-world result:  $G_N = \kappa_5^2/(6\pi L)$  for a compact extra dimension.
2. The derivation shows explicitly why a non-compact extra dimension fails (gives  $1/r^2$ ).
3. The EDC-specific power structure (12/13) does **not emerge** from this standard calculation.
4. BLOCK-003 remains **[OPEN]**: closing it requires deriving the compactification/warp scale from EDC-internal parameters.

## References

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